

## INTRODUCTION

Unpaid tuition fees are a growing financial risk for Philippine private schools. Republic Act 11984 (No Permit, No Exam Prohibition Act) bars schools from withholding examination privileges for outstanding balances, eliminating the most direct enforcement mechanism and leaving institutions dependent on proactive cash-flow management.

At the partner institution (a private school in Montalban, Rizal), **Bad Debts Expense rose from PHP 244,111 (2019) to PHP 469,102 (2025)**, with 27 students in arrears and Days Sales Outstanding (DSO) reaching 34 days, despite flexible installment plans.

Machine learning is widely applied to credit scoring, default prediction, and receivables optimization, yet existing invoice payment prediction studies share three critical gaps: (1) they use **aggregate student-level features**, discarding line-item granularity; (2) they treat payment delay as a **nominal** problem, ignoring its ordinal structure; (3) they benchmark only **5–8 classifiers**, leaving ordinal decompositions, two-stage architectures, and survival-analysis features systematically untested.

## OBJECTIVES

- Conduct a **large-scale benchmarking study** across 1,092 experimental configurations to identify the optimal model–preprocessing combination.
- Develop a **two-stage ensemble architecture** that first separates delinquent from on-time invoices, then classifies delay severity.
- Evaluate the impact of **survival-analysis-derived feature engineering** (Cox Proportional Hazards) on classifier sensitivity and recall for late-payment brackets.
- Provide an **empirically grounded framework** for private schools to transition from reactive collection practices to proactive receivables management.

## METHODOLOGY

The pipeline follows seven sequential stages, each mapped to a concrete implementation component in the project codebase. Days to Payment (DTP), the gap between due date and full settlement, is discretized into four ordinal brackets:

| Class | Label           | DTP Range  |
|-------|-----------------|------------|
| 0     | On-Time         | ≤ 0 days   |
| 1     | Slightly Late   | 1–30 days  |
| 2     | Moderately Late | 31–60 days |
| 3     | Severely Late   | 61+ days   |

### IPPP Multi-Stage Pipeline

1,092 experimental configurations

#### ① DATA ACQUISITION

Revenues File  
Itemized receivables

Enrollees File  
Installment plans

Chart of Accounts  
Category mapping

#### ② PRE-PROCESSING

Pseudonymization  
Hash student IDs  
exclude raw files

ERP Merge  
Join 3 source  
files → Fact Table

DTP Computation  
due date → payment  
date gap (days)

Target Discretization  
4 ordinal brackets  
≤0 / 1–30 / 31–60 / 61+

#### ③ FEATURE ENGINEERING (Baseline: 40 features)

Raw Financial  
4 features

DTP Historical  
11 features

Financial / Cumul.  
5 features

Behavioral  
3 features

Plan Type  
13 feats

Temporal  
4 feats

All features engineered at line-item granularity (category-level, not aggregate student balances)

#### ④ SURVIVAL ANALYSIS (Enhanced: +5 features)

Cox PH Model (lifelines)  
Fitted on censored train split  
Event = full settlement

Derived Features  
partial\_hazard · log\_partial\_hazard  
survival\_prob · cumulative\_hazard · expected\_survival

#### ⑤ CLASS BALANCING (7 strategies)

None  
baseline

SMOTE  
oversample

Borderline-SMOTE  
★ best performer  
focus on border pts

SMOTE+Tomek  
hybrid clean

Hybrid Undersample  
@0.5 / @0.7 / @0.9  
3 ratio variants

#### ⑥ MODEL TRAINING & BENCHMARKING (1,092 configs)

Base Classifiers  
AdaBoost · RF · XGBoost  
Decision Tree · GNB · KNN  
6 models  
single-stage, standard

Two-Stage Ensembles  
Stage 1 Binary (on-time vs late)  
Stage 2 Multi-class (severity)  
6 pipelines  
XGB→Ada · RF→Ada · RF→RF ...

Ordinal Classifiers  
Frank & Hall decomposition  
ordered-bracket structure  
3 models  
Ord→Ada · Ord→RF · Ord→XGB

#### ⑦ EVALUATION & DEPLOYMENT

Evaluation Metrics  
Primary: Macro-F1  
Secondary: ROC-AUC · Confusion Matrix  
Spearman ρ (F1 vs AUC) = 0.732

Dash Prototype (MVP)  
Setup Wizard · KPI Dashboard  
Prediction Drilldown · Model Compare  
Audit Logs · Settings

#### Best Configuration

TS XGBoost→AdaBoost + Borderline-SMOTE  
Macro-F1 = 0.6003 · ROC-AUC = 0.8919

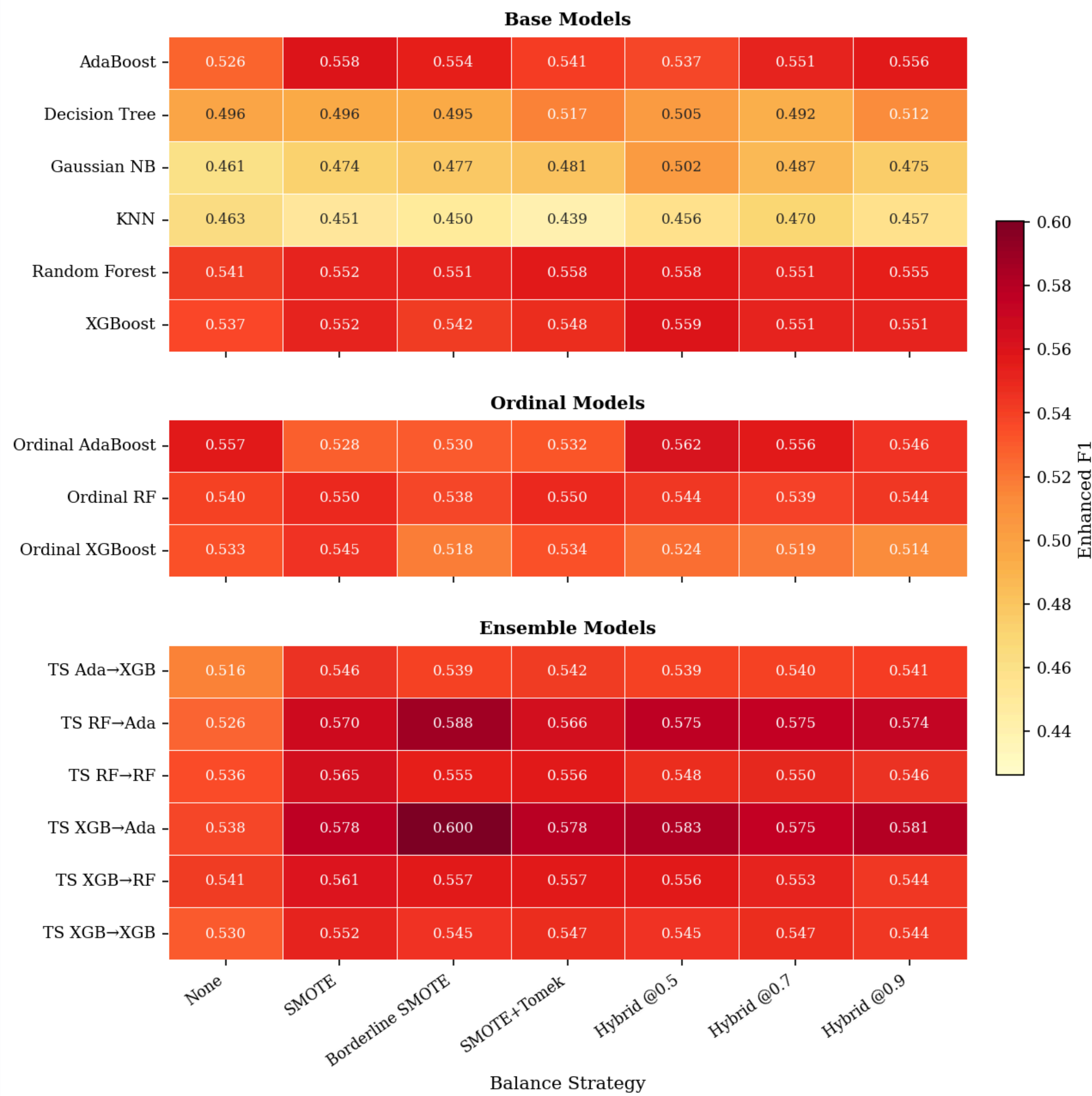
## RESULTS

### RQ1: Two-Stage vs. Ordinal vs. Base Classifiers

| Model Family | Peak F1 | Peak AUC | Best Config                   |
|--------------|---------|----------|-------------------------------|
| Two-Stage    | 0.6003  | 0.8919   | XGB→Ada + Borderline-SMOTE    |
| Ordinal      | 0.5621  | 0.8205   | Ordinal AdaBoost + Hybrid@0.5 |
| Base         | 0.5589  | 0.8607   | XGBoost + Hybrid@0.5          |

The two-stage XGBoost→AdaBoost pipeline **outperformed the best ordinal model by +0.038 F1** and the best base classifier by **+0.041 F1**. Gains are not universal across the two-stage family: hierarchical decomposition is most effective when Stage 2 pairs a high-capacity booster with a **complementary, lower-variance learner** (AdaBoost) rather than a second booster.

### F1 Macro (enhanced features) — Model × Balance Strategy



### RQ2: Survival Feature Impact

- Positive lift:** KNN (+0.0061 F1) and Gaussian Naive Bayes (+0.0050 F1): simpler models benefit from pre-computed temporal risk statistics.
- Negligible to negative:** Random Forest, XGBoost, AdaBoost: high-capacity tree ensembles already approximate survival patterns, making Cox features redundant.

Survival feature engineering has **conditional, not universal** utility: allocate effort based on the downstream learner's capacity.

### RQ3: Class-Balancing Strategy

- Borderline-SMOTE** consistently outperformed vanilla SMOTE and no-resampling baselines across all model families.
- Resampling is **essential for two-stage models**: TS XGB→Ada rises from 0.538 (none) to **0.600** (Borderline-SMOTE), a +0.062 swing.
- Hybrid undersampling** is a stable alternative: TS XGB→Ada @ Hybrid 0.5 = 0.583, only 0.017 below Borderline-SMOTE.
- Weak learners (KNN, GNB) swing most with strategy choice; strong tree ensembles (RF, XGBoost, AdaBoost) stay comparatively robust.
- Spearman ρ = 0.732 (F1 vs. AUC) confirms resampling mainly shifts **threshold placement**, not underlying class separability.

### RQ4: Feature Importance (Granular vs. Aggregate)

The three most predictive features across all families are **prev\_bracket** (most recent payment bracket), **dtp\_wavg** (recency-weighted average DTP), and **opening\_balance\_flag** (carried unpaid balance), all **behavioral-historical features only available at line-item granularity**, confirming the study's design rationale. LDA corroborates: LD1 explains 79.5% of between-class separation, and 95.9% among delinquent classes alone.

### Prototype System: Dash (Python/Plotly) Web App

|                                                                                      |                                                                                            |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Setup Wizard</b><br>Upload source files, trigger the training pipeline.           | <b>KPI Dashboard</b><br>Live F1/AUC metrics, bracket distribution, top-model ranking.      |
| <b>Prediction Drilldown</b><br>Paginated predictions, confidence scores, CSV export. | <b>Model Dashboard</b><br>ROC curves, confusion matrices, F1/AUC across all 1,092 configs. |
| <b>Audit Logs</b><br>Timestamped record of every prediction run and settings change. |                                                                                            |

## CONCLUSION

- Hierarchical architectures dominate.** Two-stage XGBoost→AdaBoost sets the performance ceiling (F1 = 0.6003, AUC = 0.8919).
- Resampling is indispensable.** With ~76% of invoices on-time, Borderline-SMOTE is essential for meaningful recall on the late-payment brackets carrying the greatest financial risk.
- Ordinal structure yields modest, consistent gains**, though less than full multi-stage decomposition delivers.
- Survival features are model-specific:** they benefit probabilistic and distance-based learners but are redundant for high-capacity tree ensembles.

A macro-F1 of 0.60 on a four-class imbalanced problem means the model correctly brackets ~60 of every 100 eventually-late invoices, with most remaining errors landing in adjacent brackets. Against zero forward visibility today, this lets finance officers **concentrate outreach on the highest-risk invoices before due dates lapse**: earlier reminders, proactive installment restructuring, and social-welfare referrals in place of after-the-fact collection that RA 11984 has rendered largely unenforceable. The framework is directly transferable to any Philippine private school sharing the same installment-billing structure, and the open-sourced prototype operationalizes the winning configuration.

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